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CALCULATION Vsl WHEN CONSTRUCTING 3D GRAPHS WITH NEGATIVE VALUES OF VARIABLES

Last year, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3]. Earlier, the author also published two articles about this problem [4, 5].

This article considers the issue of calculating the values of the lower critical volume of the Vsl spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. Vsl (GDPsl) = f(X1, X2, X3). In some cases, when the Vsl values had small minima and maxima compared to the last calculated Vsl value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Figure 1 below shows a 3D Vsl graph when the values of the variables were as follows: X1= -1...-10, X2= -1, X3= 1. Here the calculated Vsl parameter has no solutions. In the following Fig. 2 The depicted 3D Vsl graph with variables X1= -1...-10, X2= X3= -1 decreases from 0.64 to 0.02, i.e. 27.33 times.

If the variables are as follows X1= 1, X2= 5...-5, X3= -1 ...-10, then in this case the Vsl value has no solutions, as can be seen from Fig. 3. In the following Figure 4, a curve was constructed at X1 = -1, X2= 5...-5, X3 = -1 ...-10. Figure 4 shows that the Vsl parameter has a minimum of 0.003 at point 4.

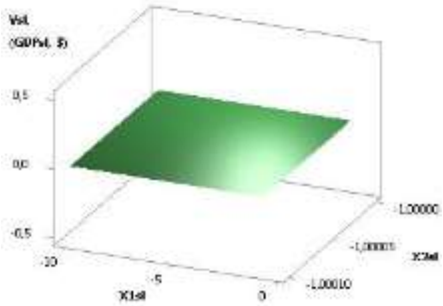

Рис. 1. 3D of Vsl (GDPsl)

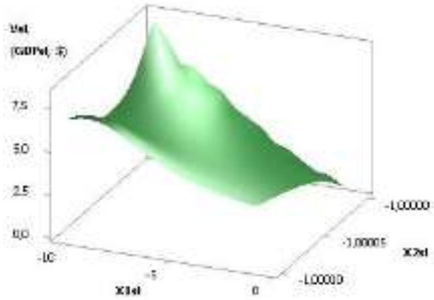

Рис. 2. 3D of Vsl (GDPsl)

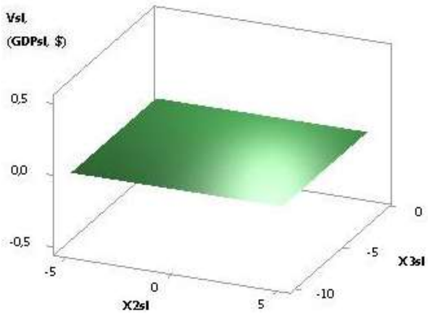

Рис. 3. 3D of Vsl (GDPsl)

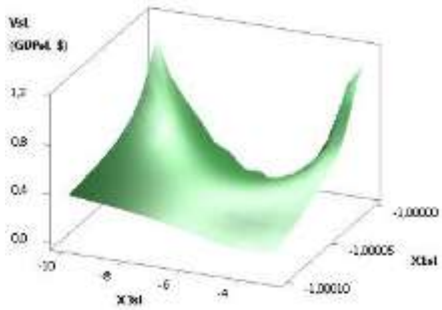

Рис. 4. 3D of Vsl (GDPsl)

The calculated values for the 3D Vsl graph in Figure 5 for variables X1= X2= 5 ...-5, X3= -1 ...-10 increases very significantly by 9485 times. From the following Figure 6 it can be seen that for variables X1 = 5 ...-5, X2= 1, X3 = -1 ...-10, the Vsl value increases slightly by 12.7 times.

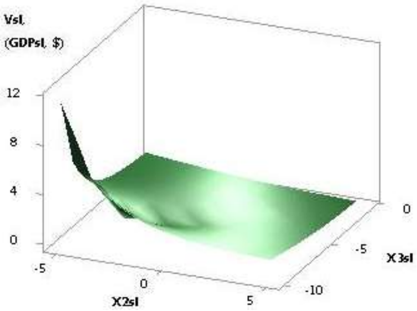

Рис. 5. 3D of Vsl (GDPsl)

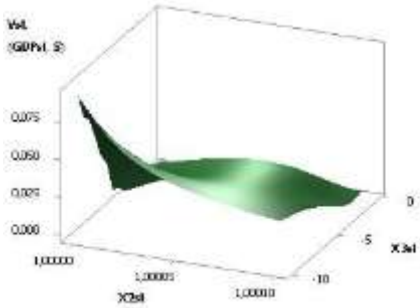

Рис. 6. 3D of Vsl (GDPsl)

Figures 7 and 8 were constructed at X1= 5...-5, X2= -1, X3= -1...-10 and X1= 1, X2= 5...-5, X3= -10...-1. Here, as in Fig. 7, the Vsl values also increase by 12.7 times, and in Fig. 8 there is no solution.

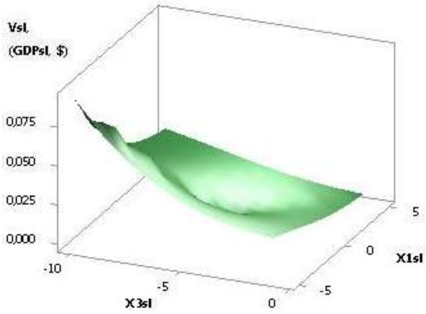

Рис. 7. 3D of Vsl (GDPsl)

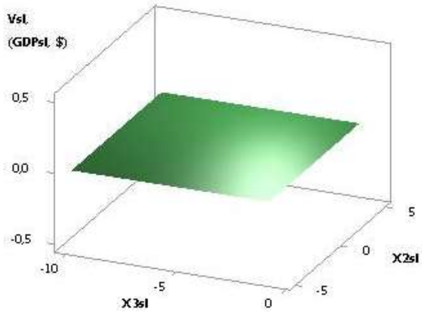

Рис. 8. 3D of Vsl (GDPsl)

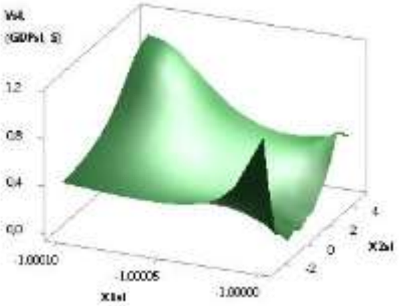

Рис. 9. 3D of Vsl (GDPsl)

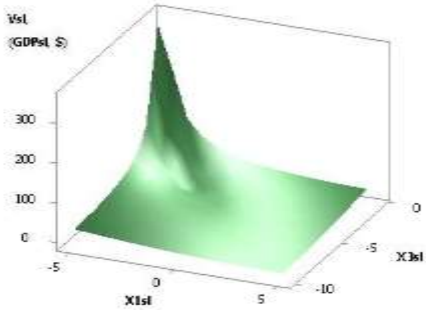

Рис. 10. 3D of Vsl (GDPsl)

In Figures 9 and 10 were constructed at X1= -1, X2= 5...-5, X3 = -10...-1 and X1= X2= 5...-5, X3= -10...-1. Here in Figure 9, the Vsl parameter has a minimum of 0.003 at point 5, and in Figure 10 it increases very significantly by 228178.84 times.

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